Combinatorics – 4

Graphs

Homework

14 сентября 2026 года

Theoretical minimum. $\sum_v \deg v = 2|E|$, so the number of odd-degree vertices is even. A graph on n vertices is a tree if and only if it is connected with $n-1$ edges, if and only if it is acyclic with $n-1$ edges, if and only if any two vertices are joined by a unique path; there are n^{n-2} labelled trees (Cayley, by the Prüfer code). A graph is bipartite exactly when it has no odd cycle; a connected graph has an Eulerian circuit exactly when all degrees are even. Hall's theorem: a bipartite graph has a matching saturating A if and only if $|N(S)| \geqslant |S|$ for all $S \subseteq A$. For a connected plane graph $v - e + f = 2$, whence $e \leqslant 3v - 6$, the non-planarity of K_5 and $K_{3,3}$, a vertex of degree at most 5, and 6-colourability. The number of walks of length k is read off from powers of the adjacency matrix, whose largest eigenvalue lies between the average and the maximum degree and whose spectrum is symmetric exactly for bipartite graphs (Linear algebra – 11).

---

1. There are n people in a city and each of them has exactly 1000 friends, friendship being symmetric. Prove that one can select a group S of people such that at least $n/2017$ of the members of S have exactly two friends inside S. (IMC)

2. Let T be a tree with n vertices and let $d(u, v)$ be the distance between u and v. Put $W(T) = \sum_{\{u,v\}} d(u, v)$ and $H(T) = \sum_{\{u,v\}} \frac{1}{d(u,v)}$, the sums being over unordered pairs of distinct vertices. Prove that $W(T)\, H(T) \geqslant \frac{(n-1)^3(n+2)}{4}$. (IMC)

3. An alien race has three genders: male, female and emale. A married triple consists of three persons, one of each gender, who all like each other; every person belongs to at most one triple, and liking is mutual.

An expedition has n members of each gender, and every member likes at least k persons of each of the two other genders. (a) Show that if n is even and $k = n/2$, it may be impossible to form even one married triple. (b) Show that if $k \geqslant 3n/4$, then a perfect system of n married triples always exists. (IMC)

4. A round-robin tournament among $2n$ teams lasted $2n-1$ days: on each day every team played one game against another team, one winning and one losing, and over the tournament each team played every other team exactly once. Can one necessarily choose one winning team from each day without choosing any team more than once? (Putnam)

5. In a tournament 2011 players meet 2011 times; every game is played by all of them at once and each player either wins or loses it. Two 2011×2011 matrices T and W start at zero; after every game, for every pair (h, k), including $h = k$, the entry T_{hk} grows by 1 if players h and k tied (both won or both lost), while if h won and k lost then W_{hk} grows by 1 and W_{kh} drops by 1. Prove that at the end $\det(T + iW)$ is a non-negative integer divisible by 2^{2010}. (Putnam 2011)

6. Let G be a simple graph on $2n$ vertices such that for every partition of its vertex set into two sets V_1 and V_2 of size n the number of edges inside V_1 equals the number of edges inside V_2. Prove that all vertices have equal degrees. (CIIM 2014)

7. Let G be a connected graph on vertices $1, \ldots, n$ with $n \geqslant 3$. For each pair of edges ij and jk sharing a vertex take the three-cycles (ijk) and (ikj) as permutations. Prove that these cycles generate the whole alternating group A_n. (Мехмат 2011)

8. Let G be a connected simple graph that is not regular, on n vertices with degrees d_v. Prove that

$$\sum_{\{u,v\}\subseteq V} (d_u - d_v)^2 \geqslant \begin{cases} n-1, & n \text{ odd}, \\ 2n-4, & n \text{ even}, \end{cases}$$

and determine when equality holds. (IMC 2025, shortlist)

9. Let G be a graph on n vertices with e edges and degrees d_v. Prove that

$$e\left(e+\binom{n-1}{2}-(n-1)\right)\geqslant(n-1)\sum_v\binom{d_v}{2},$$

and find all graphs attaining equality. (The spectral facts — $\bar{d}\leqslant\lambda_1\leqslant\Delta$, the symmetry of the spectrum for bipartite graphs, closed walks as traces — are Linear algebra – 11, homework 9.) (IMC 2025, shortlist)

10. Prove that in any graph the vertices can be split into two parts so that at least half the edges go between them, and that a graph with m edges has a bipartite subgraph with at least $\frac{m}{2}+\sqrt{\frac{m}{8}+\frac{1}{64}}-\frac{1}{8}$ edges. Compare with the probabilistic argument of Probability – 4.